

Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

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Last login: Mon Aug  3 20:23:05 on ttys000
[varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_LAB
[varnikakomali@VARNIKAs-MacBook-Air OS_LAB % cd skill
[varnikakomali@VARNIKAs-MacBook-Air skill % nano skill1.c
[varnikakomali@VARNIKAs-MacBook-Air skill % gcc skill1.c -o skill1
[varnikakomali@VARNIKAs-MacBook-Air skill % ./skill1
```

[PARENT PROCESS]

Parent PID: 22407 | Spawned Child PID: 22408

[CHILD PROCESS]

Child PID: 22408 | Parent PID: 22407

Executing 'ps -f' using execvp()...

UID	PID	PPID	C	STIME	TTY	TIME	CMD
501	22392	22391	0	8:35PM	ttys000	0:00.03	-zsh
501	22407	22392	0	8:36PM	ttys000	0:00.01	./skill1

[PARENT PROCESS] Child exited cleanly with status 0

[varnikakomali@VARNIKAS-MacBook-Air skill % ps -ej

USER	PID	PPID	PGID	SESS	JOBC	STAT	TT	TIME	COMMAND
root	1	0	1	0	0	Ss	??	144:40.95	/sbin/launc
root	523	1	523	0	0	Ss	??	37:56.29	/usr/libexe
root	524	1	524	0	0	Ss	??	0:01.26	/usr/libexe
root	525	1	525	0	0	Ss	??	3:02.08	/usr/libexe
root	527	1	527	0	0	Ss	??	18:51.37	/System/Lib
root	528	1	528	0	0	Ss	??	5:48.96	/System/Lib
root	531	1	531	0	0	Ss	??	1:54.99	/usr/sbin/s
_accessoryupdater	533	1	533	0	0	Ss	??	0:20.18	/System/Lib
_accessoryupdater	534	1	534	0	0	Ss	??	0:00.19	/usr/libexe
root	535	1	535	0	0	Ss	??	12:33.79	/usr/libexe
root	536	1	536	0	0	Ss	??	0:00.10	endpointsec
root	537	1	537	0	0	Ss	??	12:55.10	/System/Lib
root	538	1	538	0	0	Ss	??	0:00.04	/usr/libexe
root	540	1	540	0	0	Ss	??	0:05.33	/usr/libexe
root	542	1	542	0	0	Ss	??	0:00.85	/usr/libexe
root	544	1	544	0	0	Ss	??	0:02.73	/usr/libexe
_softwareupdate	545	1	545	0	0	Ss	??	0:03.15	/System/Lib
_corespeechd	546	1	546	0	0	Ss	??	0:00.03	/System/Lib
root	548	1	548	0	0	Ss	??	0:32.15	/usr/libexe
root	552	1	552	0	0	Ss	??	34:04.09	/System/Lib
_iconservices	553	1	553	0	0	Ss	??	0:02.38	/System/Lib
root	554	1	554	0	0	Ss	??	0:01.33	/usr/libexe
root	555	1	555	0	0	Ss	??	2:08.51	/usr/libexe
root	558	1	558	0	0	Ss	??	6:26.14	/usr/libexe
root	559	1	559	0	0	Ss	??	1:45.16	/usr/sbin/s
root	561	1	561	0	0	Ss	??	2:32.91	/usr/libexe
root	562	1	562	0	0	Ss	??	9:00.80	/System/Lib
root	563	1	563	0	0	Ss	??	0:14.93	/usr/libexe
root	564	1	564	0	0	Ss	??	15:01.73	/usr/libexe
root	565	1	565	0	0	Ss	??	8:33.90	/System/Lib
root	566	1	566	0	0	Ss	??	7:12.45	/System/Lib
_timed	567	1	567	0	0	Ss	??	1:13.08	/usr/libexe
_usbmuxd	568	1	568	0	0	Ss	??	0:03.86	/System/Lib
root	569	1	569	0	0	Ss	??	1:08.90	/usr/sbin/s
_locationd	571	1	571	0	0	Ss	??	45:40.80	/usr/libexe
root	572	1	572	0	0	Ss	??	0:26.67	/usr/libexe
root	574	1	574	0	0	Ss	??	0:00.09	autofs
root	575	1	575	0	0	Ss	??	17:50.28	/usr/libexe
root	576	1	576	0	0	Ss	??	0:00.07	/usr/libexe
_distnote	578	1	578	0	0	Ss	??	4:23.09	/usr/sbin/d
root	579	1	579	0	0	Ss	??	0:06.36	/System/Lib
root	581	1	581	0	0	Ss	??	0:00.16	/usr/libexe
root	582	1	582	0	0	Ss	??	0:03.01	/System/Lib
root	583	1	583	0	0	Ss	??	0:00.63	/System/Lib
root	584	1	584	0	0	Ss	??	0:00.05	/usr/sbin/K
root	585	1	585	0	0	Ss	??	0:00.03	/usr/libexe
root	586	1	586	0	0	Ss	??	0:05.77	/usr/libexe
root	589	1	589	0	0	Ss	??	7:24.23	/usr/sbin/n
root	590	1	590	0	0	Ss	??	0:19.53	/usr/libexe
root	591	1	591	0	0	Ss	??	54:31.51	/usr/libexe
root	592	1	592	0	0	Ss	??	0:41.73	/usr/libexe
root	593	1	593	0	0	Ss	??	0:34.71	/System/Lib
root	594	1	594	0	0	Ss	??	0:01.18	aslmanager
root	595	1	595	0	0	Ss	??	0:08.13	/usr/libexe
_windowserver	596	1	596	0	0	Ss	??	2490:16.69	/System/Lib
root	597	1	597	0	0	Ss	??	2:41.85	/System/Lib
varnikakomali	598	1	598	0	0	Ss	??	107:48.55	/System/Lib
root	599	1	599	0	0	Ss	??	10:04.76	/usr/sbin/c
root	600	1	600	0	0	Ss	??	0:00.68	/usr/libexe
root	601	1	601	0	0	Ss	??	0:28.35	/usr/libexe
root	602	1	602	0	0	Ss	??	0:10.94	/System/Lib
_trustd	603	1	603	0	0	Ss	??	7:13.53	/usr/libexe
_analyticsd	604	1	604	0	0	Ss	??	3:50.50	/System/Lib
root	605	1	605	0	0	Ss	??	98:46.86	/usr/libexe
_coreaudiod	606	1	606	0	0	Ss	??	217:39.67	/usr/sbin/c
root	607	1	607	0	0	Ss	??	0:00.32	/usr/libexe
root	609	1	609	0	0	Ss	??	1:56.63	/usr/libexe
root	610	1	610	0	0	Ss	??	0:16.03	/System/Lib
root	611	1	611	0	0	Ss	??	0:14.80	/usr/libexe
root	612	1	612	0	0	Ss	??	0:00.51	/System/Lib
root	613	1	613	0	0	Ss	??	1:04.85	/usr/libexe
root	614	1	614	0	0	Ss	??	0:00.06	/usr/libexe
_nsurlsessiond	615	1	615	0	0	Ss	??	5:13.22	/usr/libexe
root	616	1	616	0	0	Ss	??	0:55.34	/usr/libexe
root	617	1	617	0	0	Ss	??	0:02.55	/System/Lib
root	619	1	619	0	0	Ss	??	1:47.57	/System/Lib
root	621	1	621	0	0	Ss	??	45:12.64	/usr/libexe
root	622	1	622	0	0	Ss	??	0:34.01	/usr/libexe
root	624	1	624	0	0	Ss	??	2:24.56	/usr/libexe
_hidd	625	1	625	0	0	Ss	??	0:00.12	/usr/libexe
root	628	1	628	0	0	Ss	??	158:38.76	/usr/libexe
root	629	1	629	0	0	Ss	??	0:00.14	/usr/libexe
root	631	1	631	0	0	Ss	??	0:00.37	/usr/libexe
_biome	632	1	632	0	0	Ss	??	1:37.92	/System/Lib
_coreaudiod	634	1	634	0	1	S	??	0:32.07	/usr/sbin/d